Finite Fields and Quadratic Reciprocity Law

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F_q^* are cyclic groups

Remark: $\operatorname{ord}(a^k) = \dfrac{\operatorname{ord}(a)}{\gcd(k, \operatorname{ord}(a))}$

Lemma 1 (Order of Elements in Cyclic Groups)

Let $G = \langle g \rangle$ be a cyclic group of order n. For a divisor $d \mid n$:

1. There exists exactly one subgroup of G of order d: $H_d = \langle g^{n/d} \rangle$.
2. The number of elements of order d in G is exactly $\varphi(d)$.

1. Compute the order $\operatorname{ord}(g^{n/d}) = \frac{n}{\gcd(n/d, n)} = d$. Thus $H_d = \langle g^{n/d} \rangle$ is a subgroup of order d.
   Suppose $K \leq G$, $|K| = d$, let $m := \min\{k : a^k \in K\}$, then for $a^l \in H$,
   $$l = mq + r, \quad 0 \leq r < m$$
   Thus $r = 0$, $a^l \in \langle a^m \rangle \implies K = \langle a^m \rangle$.
2. An element $x = g^k \in G$ has order d , then $H = \langle x \rangle$ is a subgroup of order d, so $x = H_d = \langle g^{n/d} \rangle = \langle a \rangle$. Let $x = a^k$, then
   $$\operatorname{ord} x = d \iff \operatorname{ord}(a^k) = \frac{d}{\gcd(k, d)} = d$$
   This simplifies to $\gcd(k, d) = 1$. The number of such $k, (0 \leq k < d)$ is exactly $\varphi(d)$, so there are $\varphi(d)$ elements of order d.

Key Lemmas for Cyclic Groups

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$$\sum_{d|n} \varphi(d) = n$$

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Proof: For every elements $x \in G$, the order is a divisor of $|G| = n$, so

$$n = |G| = \sum_{d|n} |\{x \in G : \operatorname{ord} x = d\}| = \sum_{d|n} \varphi(d)$$

Theorem (Cyclic Group Criterion)

Let G be a finite group of order n. If for every divisor $d \mid n$, G has at most one subgroup of order d, then G is cyclic.

Proof : Let $D = \{d_1, d_2, \ldots, d_t\}$ be the set of all element orders in G.

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1. For each $d_i \in D$, the elements of order d_i are contained in the (unique) subgroup of order d_i .
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2. By Lemma 1, this subgroup has exactly $\varphi(d_i)$ elements of order d_i, so G has exactly $\varphi(d_i)$ elements of order d_i.
3. Count the total elements of G:

$$n = |G| = \sum_{i=1}^{t} \varphi(d_i)$$

By Lemma 2, $\sum_{d|n} \varphi(d) = n$. Since $D \subseteq \{d : d \mid n\}$, the equality implies $D = \{d : d \mid n\}$.

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4. Thus, $n \in D$, meaning G contains an element of order n. Hence G is cyclic.

Theorem* (Cyclic Group Criterion)

Let G be a finite ***abelian group*** of order n. If for every prime divisor $p \mid n$, G has at most one subgroup of order p, then G is cyclic.

F_q^* are cyclic groups

Exercises

Let k be a field , $f(x) \in k[x]$ such that $\deg f(x) = d$, then $f(x)$ has at most d roots in k.

Otherwise , f have roots $x_1, ..., x_m$, then

$$f(x) = (x - x_1)g(x) + r(x), \deg r < 1 \implies r(x) = 0$$

By induction $f(x) = (x - x_1)....(x - x_m)$.

Theorem

Let $\mathbb{F}_q$ be a finite field of q elements, then $\mathbb{F}_q^*$ is a cyclic group of order $q - 1$.

If $H \leq \mathbb{F}_q^*$ such that $|H| = d$, then

$$\forall x \in H, x^d = x^{|H|} = 1.$$

The polynomial $x^d - 1$ has at most d roots in $\mathbb{F}_q^*$, so $H = \{x \in \mathbb{F}_q^* : x^d = 1\}$ is uniqueness.

Legendre Symbol and Quadratic Residue Problem

Quadratic Residue Problem and Group Homomorphism

Let p be an **odd prime**.
Quadratic Residue Problem:

- Determine if the congruence $x^2 \equiv a \pmod{p}$ has a solution for $a \in \mathbb{Z}$ (with $p \nmid a$).

Key Group Homomorphism: Define the **squaring homomorphism**:

$$\sigma : \mathbb{F}_p^* \to \mathbb{F}_p^*, \quad \sigma(x) = x^2$$

1. The kernel $\ker(\sigma) = \{1, -1\}$, By the First Isomorphism Theorem:

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2. Let $H = \operatorname{Im}\sigma$, then $|H| = |\mathbb{F}_p^* / \ker(\sigma)| = \dfrac{p-1}{2}$.
3. Thus $[\mathbb{F}_p^* : H] = 2$, we know $H \lhd \mathbb{F}_p^*$. (Subgroup of index 2 must be a **normal subgroup**.)

Definition (Legendre Symbol)

The **Legendre symbol** is the map:

$$\left(\frac{\cdot}{p}\right) : \mathbb{F}_p^* \to \{1, -1\}, \quad \left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \in H \text{ (quadratic residue)}, \\ -1, & \text{if } a \notin H \text{ (quadratic non-residue)}. \end{cases}$$

For $p \mid a$, set $\left(\frac{a}{p}\right) = 0$.

Theorem 1 (Multiplicative Property of Legendre Symbol)

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \cdot \left(\frac{b}{p}\right)$$

The Legendre symbol is equivalent to the quotient group homomorphism:

$$\mathbb{F}_p^* \to \mathbb{F}_p^*/H \cong \{1, -1\}, \quad a \mapsto aH$$

$$\forall a, b \in \mathbb{F}_p^*, \quad (ab)H = (aH)(bH) \implies \left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$$

The property extends naturally to $p \mid a$ or $p \mid b$.

Euler's Criterion

Theorem 2 (Euler's Criterion)

Let p be an odd prime, $a \in \mathbb{F}_p$, then:

$$\left(\frac{a}{p}\right) = a^{(p-1)/2}.$$

1. **Case 1:** $a = 0 \in \mathbb{F}_p$, both sides equal 0.
2. **Case 2:** $a \neq 0$. Since $\mathbb{F}_p^*$ is cyclic, let g be a primitive element, so $a = g^k$ for some integer k.
3. Compute $a^{(p-1)/2} = (g^k)^{(p-1)/2} = g^{k(p-1)/2}$.
   - If a is a quadratic residue: k is even, so

$$g^{k(p-1)/2} = (g^{p-1})^{k/2} = 1^{k/2} = 1 = \left(\frac{a}{p}\right);$$

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   - If a is a quadratic residue: k is even, so

$$g^{k(p-1)/2} = (g^{p-1})^{k/2} = 1^{k/2} = 1 = \left(\frac{a}{p}\right);$$

   - If a is a quadratic non-residue: since $g^{p-1} = 1$ and $g^{(p-1)/2} \neq 1$ and k is odd, so

$$g^{k(p-1)/2} = g^{(p-1)/2} = -1 = \left(\frac{a}{p}\right).$$

Field extension

Character and extension of field

character of fields

Let K be a field, $\mathcal{P}$ be the prime field, then define $\operatorname{char} K = \begin{cases} 0 & \text{if } \mathcal{P} \simeq \mathbb{Q} \\ p & \text{if } \mathcal{P} \simeq \mathbb{F}_p \end{cases}$

extension of field

If L, K are fields. An extiension is an injective morphism $K \to L$ as rings, then we say L be an extension field. And we write L/K is an extenstion.

proposition

1. If L is an extension field of K, then $\mathrm{char} L = \mathrm{char} K$.
2. If K is a field of $\mathrm{char} K = p$, then $\forall x_1, ..., x_m \in K$

$$(x_1 + x_2)^p = x_1^p + x_2^p;$$

$$(x_1 + ... + x_m)^p = x_1^p + ... + x_m^p;$$

the map $K \to K,\ u \mapsto u^p$ is an isomorphism, called Frobenius isomorphism.

field extension

proposition

If L/K is an extension of field, then L is a vector space over K. The dimension $\dim_K L$ is called the degree of extension L/K, denoted by $[L:K]$.

proposition

If $p(x) \in K[x]$ is irreducible, then $L = K[x]/(p(x))$ is an extension field of K. Let $z = x + (p(x)) \in K[x]/(p(x))$, then

$$1, z, ..., z^{\deg p - 1}$$

form a basis of L over K, thus $[L:K] = \deg p$.

Proof: If there are $c_0, ..., c_{d-1} \in K \to L$ such that

$$c_0 + c_1 z + ... + c_d z^{d-1} = 0$$

contradicts to $p(x)$ is the minimal polynomial of z.

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Example

$[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = n$.

Theorem

$L/K, E/L$ be finite extension, then $[E:K] = [E:L][L:K]$.

Proof: Write on blackboard.

Theorem (splitting field)

For nonconstant $f(x) \in K[x]$

1. there is an extension of field L/K, such that $f(x) \in K[x] \subseteq L[x]$ split in L. (i.e. L contain all roots of $f(x)$.)
2. $[L:K] < \infty$.

Example

$\mathbb{Q}(\sqrt[3]{2}, i)$ be an splitting field of $x^3 - 2 \in \mathbb{Q}[x]$.

Example

The splitting field of $\mathbb{F}_q$ is a finite field.

Primitive p-th root in finite field

Let p, q be distinct odd primes.
Primitive p-th root of unity is an element such that $\mathrm{ord} = p$.

Lemma

There is a primitive p-th root of unity some **extension field** of $\mathbb{F}_q$.

1. Let E be the **splitting field** of $f(x) = x^p - 1 \in \mathbb{F}_q[x]$ over field $\mathbb{F}_q$, then E be a finite field over $\mathbb{F}_q$.
2. Since $f'(x) = px^{p-1} \neq 0, (f, f') = 1$, $f(x)$ have no repeat roots, so all the roots
$$H = \{x \in E : x^p - 1 = 0\}$$
form a subgroup of E^*.
3. Then $|H| = p$, so $p||E^*|$, thus there exists $\zeta_p \in E^*$ of order p. (Cauchy theorem)

Quadratic Reciprocity Law

Statement of Quadratic Reciprocity Law

Theorem 3 (Gauss's Quadratic Reciprocity Law)

Let p and q be **distinct odd primes**, then:

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}.$$

Equivalent Formulation

- If either $p \equiv 1 \pmod 4$ or $q \equiv 1 \pmod 4$, then $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$.
- If both $p \equiv 3 \pmod 4$ and $q \equiv 3 \pmod 4$, then $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$.

Example

$$\left(\frac{17}{97}\right)\left(\frac{97}{17}\right) = (-1)^{\frac{17-1}{2}\frac{97-1}{2}} = 1 \Rightarrow$$

$$\left(\frac{17}{97}\right) = \left(\frac{97}{17}\right) = \left(\frac{12}{17}\right) = \left(\frac{2^2}{17}\right)\left(\frac{3}{17}\right) = \left(\frac{3}{17}\right) = \left(\frac{17}{3}\right) = \left(\frac{2}{3}\right) = -1$$

Proof of Quadratic Reciprocity Law

Definition 3 (Gauss Sum)

Define the Gauss sum associated with (p, q) is

$$G(p,q) := \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta_p^a = \sum_{a \in \mathbb{F}_p^*} \left(\frac{a}{p}\right) \zeta_p^a \in E.$$

Key Lemmas for Gauss Sum

1. $G^2 = (-1)^{\frac{p-1}{2}} p$;
2. $G^q = \left(\frac{q}{p}\right) G$.

Final Step

- Compute $G^{q-1} = \left(\frac{q}{p}\right)$. On the other hand,
  $G^{q-1} = \left(G^2\right)^{\frac{q-1}{2}} = p^{\frac{q-1}{2}} (-1)^{\frac{(p-1)(q-1)}{4}}$.
- By Euler's criterion, $p^{\frac{q-1}{2}} = \left(\frac{p}{q}\right)$. Thus:

$$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right) (-1)^{\frac{(p-1)(q-1)}{4}} \implies \left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}.$$

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Propsition

1. $G^q = \left(\frac{q}{p}\right) G$;
2. $G^2 = (-1)^{\frac{p-1}{2}} p$.

$$G^q = \sum_{a \in \mathbb{F}_p^*} \left(\frac{a}{p}\right)^q \zeta_p^{aq} \overset{k=aq}{=\!=} \sum_{k \in \mathbb{F}_p^*} \left(\frac{kq^{-1}}{p}\right) \zeta_p^k$$

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$$= \sum_{k \in \mathbb{F}_p^*} \left(\frac{q^2}{p}\right) \left(\frac{kq^{-1}}{p}\right) \zeta_p^k = \sum_{k \in \mathbb{F}_p^*} \left(\frac{kq}{p}\right) \zeta_p^k$$

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1. $G^q = \left(\frac{q}{p}\right) G$;
2. $G^2 = (-1)^{\frac{p-1}{2}} p$.

$$\begin{aligned} G^q &= \sum_{a \in \mathbb{F}_p^*} \left(\frac{a}{p}\right)^q \zeta_p^{aq} \overset{k=aq}{=\!=} \sum_{k \in \mathbb{F}_p^*} \left(\frac{kq^{-1}}{p}\right) \zeta_p^k \\ &= \sum_{k \in \mathbb{F}_p^*} \left(\frac{q^2}{p}\right) \left(\frac{kq^{-1}}{p}\right) \zeta_p^k = \sum_{k \in \mathbb{F}_p^*} \left(\frac{kq}{p}\right) \zeta_p^k \\ &= \left(\frac{q}{p}\right) \sum_{k \in \mathbb{F}_p^*} \left(\frac{k}{p}\right) \zeta_p^k = \left(\frac{q}{p}\right) G \end{aligned}$$

$$G(p,q)^2 = \sum_{a\in\mathbb{F}_p^*}\sum_{b\in\mathbb{F}_p^*}\left(\frac{a}{p}\right)\left(\frac{b}{p}\right)\zeta_p^{a+b}$$

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$$\overset{k=ba^{-1}}{=\!=} \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{a}{p}\right)\left(\frac{ak}{p}\right)\zeta_p^{a+ak} = \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{a^2}{p}\right)\left(\frac{k}{p}\right)\zeta_p^{a(1+k)}$$

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\begin{aligned}
G(p,q)^2 &= \sum_{a\in\mathbb{F}_p^*}\sum_{b\in\mathbb{F}_p^*}\left(\frac{a}{p}\right)\left(\frac{b}{p}\right)\zeta_p^{a+b} \\
&\overset{k=ba^{-1}}{=\!=} \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{a}{p}\right)\left(\frac{ak}{p}\right)\zeta_p^{a+ak} = \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{a^2}{p}\right)\left(\frac{k}{p}\right)\zeta_p^{a(1+k)} \\
&= \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\zeta_p^{a(1+k)} = \sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\sum_{a\in\mathbb{F}_p^*}\left(\zeta_p^{1+k}\right)^a
\end{aligned}
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&= \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\zeta_p^{a(1+k)} = \sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\sum_{a\in\mathbb{F}_p^*}\left(\zeta_p^{1+k}\right)^a \\
&= \sum_{1+k=0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}1^a + \sum_{1+k\neq 0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}\left(\zeta_p^{1+k}\right)^a
\end{aligned}
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&= \sum_{1+k=0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}1^a + \sum_{1+k\neq 0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}\left(\zeta_p^{1+k}\right)^a \\
&= \left(\frac{-1}{p}\right)(p-1) + \sum_{k\neq -1}\left(\frac{k}{p}\right)\left(-1+\sum_{a=0}^{p-1}\left(\zeta_p^{1+k}\right)^a\right)
\end{aligned}
$$

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\begin{aligned}
G(p,q)^2 &= \sum_{a\in\mathbb{F}_p^*}\sum_{b\in\mathbb{F}_p^*}\left(\frac{a}{p}\right)\left(\frac{b}{p}\right)\zeta_p^{a+b} \\
&\overset{k=ba^{-1}}{=\!=} \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{a}{p}\right)\left(\frac{ak}{p}\right)\zeta_p^{a+ak} = \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{a^2}{p}\right)\left(\frac{k}{p}\right)\zeta_p^{a(1+k)} \\
&= \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\zeta_p^{a(1+k)} = \sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\sum_{a\in\mathbb{F}_p^*}\left(\zeta_p^{1+k}\right)^a \\
&= \sum_{1+k=0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}1^a + \sum_{1+k\neq 0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}\left(\zeta_p^{1+k}\right)^a \\
&= \left(\frac{-1}{p}\right)(p-1) + \sum_{k\neq -1}\left(\frac{k}{p}\right)\left(-1+\sum_{a=0}^{p-1}\left(\zeta_p^{1+k}\right)^a\right) \\
&= \left(\frac{-1}{p}\right)(p-1) + \sum_{k\neq -1}\left(\frac{k}{p}\right)(-1)
\end{aligned}
$$

$$
\begin{aligned}
G(p,q)^2 &= \sum_{a\in\mathbb{F}_p^*}\sum_{b\in\mathbb{F}_p^*}\left(\frac{a}{p}\right)\left(\frac{b}{p}\right)\zeta_p^{a+b} \\
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&= \sum_{a\in\mathbb{F}_p^*}\sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\zeta_p^{a(1+k)} = \sum_{k\in\mathbb{F}_p^*}\left(\frac{k}{p}\right)\sum_{a\in\mathbb{F}_p^*}\left(\zeta_p^{1+k}\right)^a \\
&= \sum_{1+k=0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}1^a + \sum_{1+k\neq 0}\left(\frac{k}{p}\right)\sum_{a=1}^{p-1}\left(\zeta_p^{1+k}\right)^a \\
&= \left(\frac{-1}{p}\right)(p-1) + \sum_{k\neq -1}\left(\frac{k}{p}\right)\left(-1+\sum_{a=0}^{p-1}\left(\zeta_p^{1+k}\right)^a\right) \\
&= \left(\frac{-1}{p}\right)(p-1) + \sum_{k\neq -1}\left(\frac{k}{p}\right)(-1) \\
&= \left(\frac{-1}{p}\right)(p-1) - \sum_{k}\left(\frac{k}{p}\right) + \left(\frac{-1}{p}\right) = \left(\frac{-1}{p}\right)p \in E.
\end{aligned}
$$

Application

Theorem1

For odd prime p, $a \in \mathbb{F}_p$, then:

$$\left(\frac{a}{p}\right) = a^{(p-1)/2}.$$

Theorem 2 (Gauss's Quadratic Reciprocity Law)

Let p and q be **distinct odd primes**, then:

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}.$$

Theorem3

For odd prime p, we have

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

Proof of theorem 3

Let ζ be 8-th ,then

$$\zeta^8 = 1 \Rightarrow \zeta^4 = -1$$

Let $y := \zeta + \zeta^{-1}$, then

$$y^2 - 2 = \left(\zeta + \zeta^{-1}\right)^2 - 2 = \zeta^2 + \zeta^{-2} = \zeta^{-2}\left(1 + \zeta^4\right) = 0$$

So by Euler fomula

$$\left(\frac{2}{p}\right) = 2^{\frac{p-1}{2}} = \left(y^2\right)^{\frac{p-1}{2}} = y^{p-1}$$

- If $p \equiv \pm 1 \pmod 8$, then

$$y^p = \left(\zeta + \zeta^{-1}\right)^p = \zeta^p + \zeta^{-p} = \zeta + \zeta^{-1} = y$$

- If $p \equiv \pm 5 \pmod 8$, then

$$y^p = \left(\zeta + \zeta^{-1}\right)^p = \zeta^p + \zeta^{-p} = \zeta^5 + \zeta^{-5} = -\zeta - \zeta^{-1} = -y$$

Primitive n-th root in finite field

Let p be odd primes.
Primitive n-th root of unity is an element such that $\mathrm{ord} = n$.

Lemma

If $(n, p) = 1$, then there is a primitive n-th root of unity some **extension field** of $\mathbb{F}_p$.

1. Let E be the **splitting field** of $f(x) = x^n - 1 \in \mathbb{F}_p[x]$ over field $\mathbb{F}_p$, then E be a finite field over $\mathbb{F}_p$.
2. Since $f'(x) = nx^{n-1} \neq 0, (f, f') = 1$, $f(x)$ have no repeat roots, so all the roots
$$H = \{x \in E : x^n - 1 = 0\}$$
form a subgroup of E^*.
3. Then $|H| = n$ and it is a cyclic group, thus there exists $\zeta_n \in H$ of order n.

Execises

Is the equation has a solution in $\mathbb{Z}$?

- $x^2 + 99x - 16 \equiv 0 \bmod 97$
- $x^2 \equiv 2 \bmod 97$

All errors and omissions are my own responsibility.
以上一切谬误不当之处皆属本人之咎.

Any Comments Are Welcome!
望赐教!